

1. Solve the following problem

Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
(b) Find $P(0 < X \leq 1)$.
(c) Find the Cumulative density function and use it to evaluate $P(0 < X \leq 1)$.

2. Solve the following problem

A continuous RV has a pdf $f(x) = 3x^2$, $0 \leq x \leq 1$. Find a and b such that

- (a) $P(X \leq a) = P(X > a)$, and
(b) $P(X > b) = 0.05$.

3. Solve the following problem

If the density function of a continuous RV X is given by

$$\begin{aligned} f(x) &= ax, & 0 \leq x \leq 1 \\ &= a, & 1 \leq x \leq 2 \\ &= 3a - ax, & 2 \leq x \leq 3 \\ &= 0, & \text{elsewhere} \end{aligned}$$

- (a) find the value of a , and
(b) find the cdf of X .
(c) If x_1, x_2 and x_3 are 3 independent observations of X , what is the probability that exactly one of these 3 is greater than 1.5?

4. Solve the following problem

A continuous RV X that can assume any value between $x = 2$ and $x = 5$ has a density function given by $f(x) = k(1 + x)$. Find $P(X < 4)$.

5. Solve the following problem

The distribution function of a RV X is given by $F(x) = 1 - (1 + x)e^{-x}$, $x \geq 0$. Find the density function, mean and variance of X .

6. Solve the following problem

Q: The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X that has the density function

$$f(x) = \begin{cases} \frac{2(x+2)}{5}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Show that $P(0 < X < 1) = 1$.
(b) Find the probability that more than 1/4 but fewer than 1/2 of the people contacted will respond to this type of solicitation.

7. Solve the following problem

: Suppose a certain type of small data processing firm is so specialized that some have difficulty making a profit in their first year of operation. The probability density function that characterizes the proportion Y that make a profit is given by

$$f(y) = \begin{cases} ky^4(1-y)^3, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) What is the value of k that renders the above a valid density function?
- (b) Find the probability that at most 50% of the firms make a profit in the first year.
- (c) Find the probability that at least 80% of the firms make a profit in the first year.

8. Solve the following problem

: The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.